The intensity boxplot

The same distributions as the histogram, summarised so that every condition — or every sample — fits on one axis.

AT A GLANCE

Where	Quality control → Boxplot
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation. The conditions, and which samples belong to each, come from the annotation.
Data source	The data as loaded, on the $\log_2$ scale. There is no selector: quality control describes what the instrument produced, before filtering or imputation.
Two modes	**Type** Mean draws one box per condition; **Type** Single draws one box per sample.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

A boxplot answers the same question as the histogram — do the samples sit on one scale — for far more samples at once. Twenty overlaid density curves are unreadable; twenty boxes are not, and a single displaced median is found faster here than anywhere else in the tool. What it costs is shape: a box states five numbers, so two distributions that differ everywhere except in their quartiles draw the same box. Use this page to find the sample worth looking at, then

the histogram to see what is wrong with it.

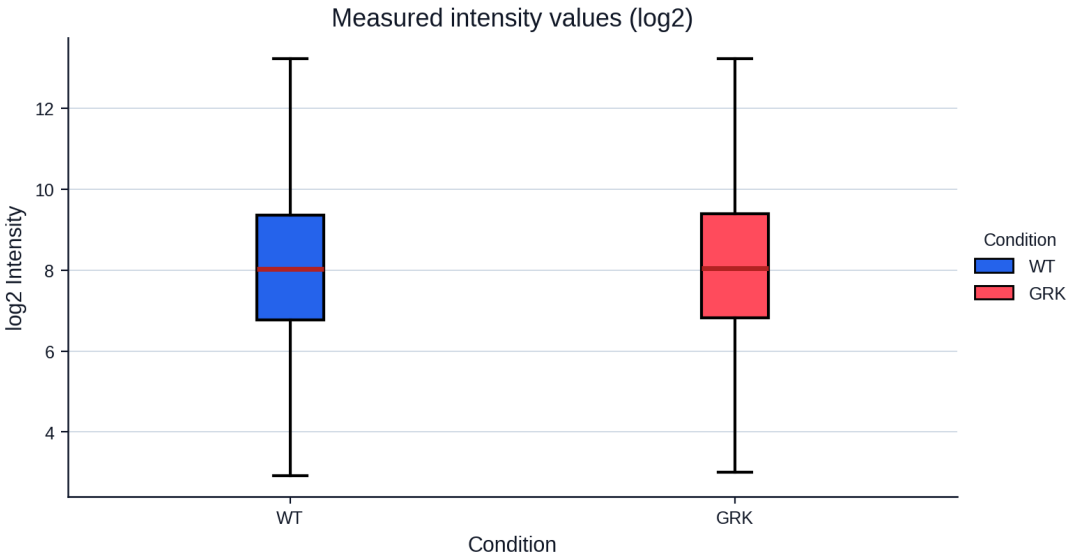


Figure 1.1. Type Mean at protein level on the example dataset: one box per condition, built from the condition means of the five WT and the five GRK samples. The two boxes sit at the same median and span the same range, which is the reading this page exists to give.

Reading it

- The box: the interquartile range, from the 25th to the 75th percentile. Its fill is the condition colour.
- The line across the box: the median. It is drawn heavier and in its own colour so it reads at a glance across many boxes.
- The whiskers: the most extreme values still inside 1.5 × IQR of the box. The caps are data points, not the limits themselves.
- The outlier markers: everything beyond the whiskers, drawn only when **Toggle Outliers** is on.
- The label under each box: the condition in Mean mode, and condition_n in Single mode — numbered within the condition rather than across the experiment, with the original sample identifier added in brackets when **Toggle IDs** is on.

> **EASY TO GET WRONG**
>
> Two very different distributions can draw the same box. A bimodal sample and a unimodal one with the same quartiles are indistinguishable here, and so are a symmetric distribution and a skewed one whose median happens to sit centrally. Whenever a box looks acceptable but a downstream result does not, open the histogram for the same level: it is the same data, drawn without the summary.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Type	Mean: one box per condition, built from the condition-wise mean of each feature. Single: one box per sample, built from every value in that column.	Mean	page
Toggle Outliers	Draws the points beyond the whiskers. With it off the whiskers are still computed the same way; the points outside them are simply not marked.	off	page
Toggle IDs	Adds the original sample identifier in brackets after each sample label. Single mode only — in Mean mode the labels are conditions and the switch does nothing. It does not change the table.	off	page
Toggle Text	Prints the median above each box, to two decimals.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Box border color	The outline of every box, and of its whiskers and caps.	black	appearance
Median color	The line across the box.	firebrick	appearance
Outlier dot color, Outlier outline color, Outlier size (pt)	The markers beyond the whiskers. The three rows appear only while **Toggle Outliers** is on.	red, black, 5	appearance
Value labels	Size of the number printed on each box by **Toggle Text**. Until it is set the plot uses 9.	9	appearance
Condition names, Condition colors	One display name and one colour per condition; the colour is the box fill. In Single mode the sample names can be overridden as well.	the condition palette	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size, weight and colour. The title is written by the plot: Measured intensity values (log2).	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

The box, the whiskers and what counts as an outlier follow the standard definition given in **Concepts**; it is not repeated here. What differs between the two modes is only what goes into the box.

MEAN MODE

The matrix is the data as loaded, on the $\log_2$ scale; non-numeric and non-finite entries are NA. For every feature i and condition c the mean over the replicates that carry a value is taken,

$$\bar{x}_{i,c} = \frac{1}{|O_{i,c}|} \sum_{j \in O_{i,c}} x_{ij}, \qquad O_{i,c} = \{\, j \in S_c \,:\, x_{ij} \neq \mathrm{NA} \,\},$$

and the box of condition c is built from the vector $\{\bar{x}_{i,c}\}$ — one value per feature. Averaging first removes the sample-to-sample scatter, so what remains is the shape of the condition.

SINGLE MODE

No averaging. The box of a sample is built from every non-missing value in that column, one per feature. This is the distribution the instrument actually produced for that run, and it is the mode to use when looking for one bad sample inside a set of

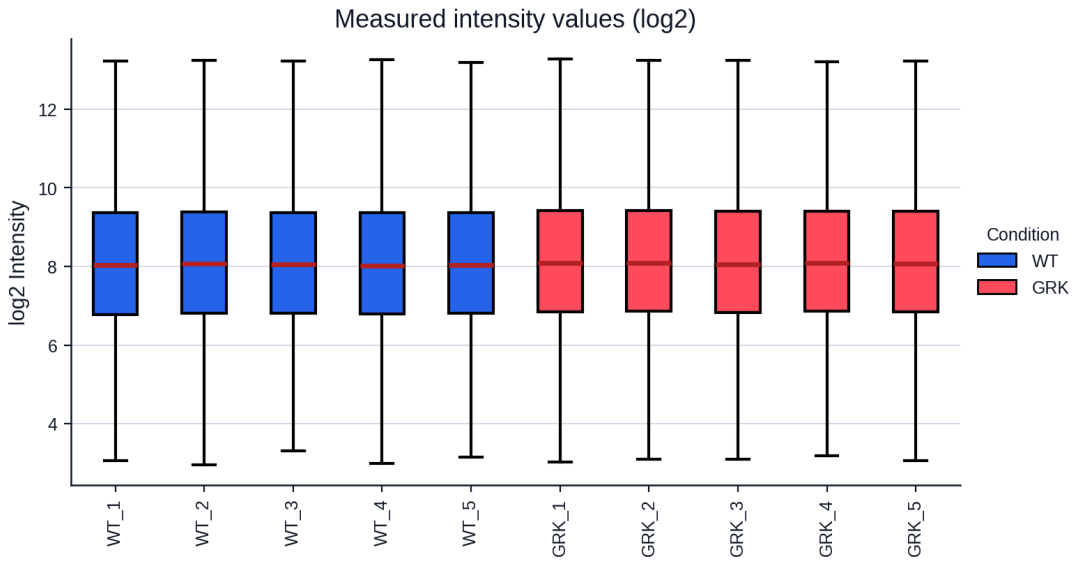


Figure 1.2. Single mode on the same data: ten boxes, one per sample, labelled WT_1 to GRK_5 and grouped by condition. Nothing is averaged away, and every box still sits at the same median — the evenness the summary in Mean mode could have hidden.

replicates.
Samples are relabelled condition_n, numbered within the condition in the order the annotation gives, and the boxes are drawn grouped by condition rather than in file order.

Compare samples with similar roles in the design. A low median or unusually wide box can reflect loading, instrument response, biological composition, missingness or processing; the plot alone does not decide which. Check Single mode before Mean, since averaging across replicates can hide an unusual sample. If normalisation aligned the distributions, their similarity is partly a result of that operation, not independent validation.

Edge cases

- In Mean mode a condition whose features are all missing yields no finite means and is left out of the axis entirely. In Single mode a sample with no values keeps its position on the axis, so the gap is visible.
- A feature missing in some replicates of a condition is averaged over the ones it has, so in Mean mode the boxes of different conditions can rest on different numbers of features.
- A condition with one replicate is legitimate here: in Mean mode its box is that sample's own distribution.
- **Toggle Text** draws the number at 1.02 times the median, so on a negative median — possible at phosphoprotein level after a robust_site_median collapse — the label sits just below the line rather than above it.
- At phosphoprotein level collapsed with robust_site_median the y axis reads log2 Ratio to Site Median: those values are ratios centred on each site's own median, and comparing

their absolute height with an intensity is meaningless.

4 The table below the plot

The table reports the numbers that draw the boxes, each rounded to two decimals. Its first column follows the mode: one row per condition in Mean mode, one row per sample in Single mode.

Columns in this table

Column	What it holds
Condition	Mean mode. The condition, as the annotation names it.
Sample	Single mode. The sample, labelled condition_n as on the plot. **Toggle IDs** does not change this column.
mean	The arithmetic mean of the values in the box. It is not drawn on the plot, and it is not the median.
median	The line across the box.
Q1	The lower edge of the box, the 25th percentile.
Q3	The upper edge of the box, the 75th percentile.
lower_whisker	The smallest value at or above $Q1 - 1.5 \times \text{IQR}$ — where the lower whisker ends.
upper_whisker	The largest value at or below $Q3 + 1.5 \times \text{IQR}$ — where the upper whisker ends.

The two whisker columns are data values, not the cutoffs themselves, which is why they move in steps rather than continuously as a sample changes.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

```python
fig = boxplot_int(data, meta)
fig = boxplot_int_single(data, meta, id=False)
```

```r
fig <- boxplot_int(data, meta)
fig <- boxplot_int_single(data, meta, id = FALSE)
```

The two modes are two functions rather than an argument: boxplot_int is **Type** Mean and boxplot_int_single is Single. outliers=True is **Toggle Outliers**, text=True is **Toggle Text**, and id — which only the single-sample function takes, and which defaults to True there rather than to off as on the page — adds the original identifier in brackets.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

7 Further reading

NIST: box plots introduces quartiles, whiskers and observations beyond the whiskers. Such observations are not automatically measurement errors.

Behind this page

Function	What it produces
boxplot_int	One box per condition, from the condition-wise mean $\log_2$ intensities
boxplot_int_single	One box per sample, from every value in the column
prepare_boxplot_df	The condition-level table
prepare_boxplot_single_df	The per-sample table